

Mutant epidemics in discrete structures

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Abstract: We will introduce the SIS and SIR epidemic models on complex networks and study the effect of mutations on the evolution of an epidemic. To do that, we will run simulations on a real-world complex network and study if its behaviour adjusts to the SIS or the SIR model depending on the value of the mutation rate. With these data, we will provide a phase diagram for an epidemic outbreak.

Keywords: Complex Networks, stochastic processes, epidemic spreading.

SDGs: 3.3, 3.D.

I. INTRODUCTION

Complex networks are a powerful tool to model population dynamics and, in particular, epidemic outbreaks. We can represent a population using a complex network in which each vertex represents an individual and the edges represent physical interactions between them, that is, possible ways of propagation of a disease. There exist two main models of epidemic dynamics, the SIR and SIS, that differ in the presence or lack of immunity in recovered individuals. Now, if we consider the possibility that viruses can mutate, the dynamics of this epidemic will be somewhere halfway the SIR and SIS models, because recovered individuals will acquire immunity to a certain strain but can still be infected by others.

In section II we will explain the SIR, SIS and mutations models and provide some simulations of a mutant epidemic outbreak on a real-world complex network to appreciate the behaviour of the epidemic depending on its parameters. In section III we will develop this problem by computing the phase diagram of an epidemic, that is, a division of the phase space into three parts. There will be points of the phase space for which the epidemic does not thrive and dies almost immediately. For the other two fractions of the phase space, the points in one part correspond to parameters for which the epidemic adjusts to an SIR model, and in the other one the points adjust to an SIS model.

II. EPIDEMIC MODELS

In this section, we will present epidemic models that consider homogeneous uncorrelated networks [1], that is, networks in which each vertex has approximately the same number of neighbours, $k_i \simeq \langle k \rangle \forall i$. We will also assume the homogeneous mixing hypothesis [2], that states that the rate of acquisition of the disease is proportional to the density of infected individuals.

A. The SIR model

SIR stands for Susceptible-Infected-Removed. These are the three possible states in which individuals can be in this model. Susceptible individuals can become infected with a certain probability, which is given by the infection rate, λ , when at least one of their neighbours is infected. On the other hand, infected individuals can recover and gain immunity to the disease or, in the worst case, die at any moment with a probability given by the infection rate δ . In both cases, these individuals can no longer affect the development of the epidemic. The dimensions of this rates are individuals per unit of time, and usually δ is taken as 1 and λ is transformed to $\frac{\lambda}{\delta}$ to simplify the calculations. Then, the results can be rescaled to adjust to the real rates. So, from now on we will express the time in units of inverse of δ and all the rates in units of δ .

The variables we will be working with are the fraction of susceptible, infected, and recovered individuals, written as $S(t)$, $\rho(t)$ and $R(t)$, respectively. Since each individual must be in only one of these states, we have that

$$S(t) + \rho(t) + R(t) = 1 \quad (1)$$

And the equations that define the evolution of the epidemic are

$$\frac{dS}{dt} = -\lambda \langle k \rangle \rho S \quad (2)$$

$$\frac{d\rho}{dt} = -\rho + \lambda \langle k \rangle \rho S \quad (3)$$

$$\frac{dR}{dt} = \rho \quad (4)$$

What these equations mean is that infected individuals recover at a unity rate (equation (4)) and the velocity of infection for susceptible individuals is proportional to the densities of infected and susceptible individuals, to λ and to the mean degree $\langle k \rangle$ (equation (2)), which represents the number of contacts per unit of time for each individual. Notice that by substituting equation (4) in (2) we obtain the differential equation $\frac{dS}{S} = -\lambda \langle k \rangle dR$.

If we suppose that at the start of the outbreak all the population is susceptible to the virus, that is, $S(0) \simeq 1$, the solution of this equations is

$$S = e^{-\lambda \langle k \rangle R(t)} \quad (5)$$

In the SIR model, the epidemic prevalence is defined as the number of infected individuals in the whole epidemic process. This is equal to the number of recovered individuals at $t = \infty$, so we will write prevalence as R_∞ .

Now, if we take into account that at $t = \infty$ there will be no infected individuals, using equation (1) we get that

$$R_\infty = 1 - S_\infty = 1 - e^{-\lambda \langle k \rangle R_\infty} \quad (6)$$

This equation will always have the trivial solution $R_\infty = 0$, which means that the epidemic dies immediately. But there exists a condition in order to have a nonzero solution, which implies that the epidemic affects a macroscopic fraction of the population. Since the functions $f(R_\infty) = R_\infty$ and $g(R_\infty) = 1 - e^{-\lambda \langle k \rangle R_\infty}$ take the same value at $R_\infty = 0$ and $g(R_\infty)$ is a concave function, the equation $R_\infty = g(R_\infty)$ is satisfied at a nonzero value if and only if

$$\frac{d}{dR_\infty}(1 - e^{-\lambda \langle k \rangle R_\infty})|_{R_\infty=0} \geq 1. \quad (7)$$

This condition is satisfied when $\lambda \geq \langle k \rangle^{-1}$. So we can define a critical value of the infection rate, $\lambda_c = \langle k \rangle^{-1}$, which we will call the epidemic threshold, and it will be the minimum value that λ has to take to prevent the epidemic from immediately disappearing.

B. The SIS model

SIS stands for susceptible-infected-susceptible. In this model, recovered individuals do not acquire immunity to the disease, in contrast with the SIR model, so they can be infected again. In this case, individuals alternate between the two possible states following a stochastic cycle, and we only need one equation to describe the epidemic dynamics, equivalent to equation (3):

$$\frac{d\rho(t)}{dt} = -\rho(t) + \lambda \langle k \rangle (1 - \rho(t)) \quad (8)$$

It can be proven [3] that, as in the SIR model, there exists an epidemic threshold, $\lambda_c = \langle k \rangle^{-1}$, such that for $\lambda \leq \lambda_c$ the epidemic dies exponentially fast, and for $\lambda > \lambda_c$ there is always a macroscopic fraction of the population infected. If this happens, we say that the epidemic has arrived to an endemic phase.

C. A model with mutations

The model we will consider is similar to the SIR, with the addition that infected individuals can host mutations.

That is, when the virus that causes the disease replicates inside the body of an infected individual, there is a nonzero probability, which will be given by the mutation rate γ , that its genetic code suffers an alteration in the process, which can create a new strain of this virus. Recovered individuals, who have acquired immunity for the original virus, may be susceptible to this new strain and can become infected again. As we have done with λ , we will express the rate of mutation in units of δ .

Taking this into account, we have a set of equations similar to the ones of the SIR (equations (2), (3) and (4)) for every strain alive at time t . That is, we have to consider an stochastic number of stochastic processes, so the analytical treatment of this problem, if possible, can be very tedious and complicated.

Depending on the value of the parameter γ , the epidemic dynamics will resemble the ones in an SIR model, that is, $\rho_\infty = 0$, or behave like a SIS model, arriving to an endemic phase and never totally extinguishing. To expose this dependence, we have plotted in figure 1 two simulations with different values of γ . In one simulation, the density of infected individuals oscillates for a while and then goes to 0, so the epidemic extinguishes as in an SIR model. In the other case, after a certain time, ρ arrives to a stable nonzero density and then oscillates slightly around this value permanently, as it happens in an SIS model.

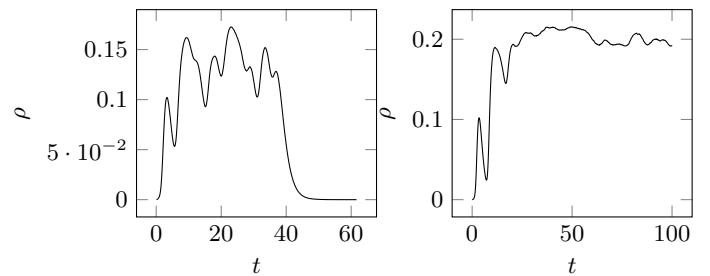


FIG. 1: Time evolution of the fraction of infected individuals in two simulations of an epidemic outbreak on a complex network. In both cases the rate of infection is $\lambda = 0.1$. The values for the mutation rate are $N\gamma = 9.28$ for the graphic on the left and $N\gamma = 26.52$ for the graphic on the right. The time is in units of δ^{-1} .

D. Computational simulation of a mutant epidemic

All the results exposed in this thesis have been obtained via computational simulations. To simulate a mutant epidemic, we have adapted the Gillespie algorithm [4]. We have considered that the time of recovery for an infected individual, the time in which a mutation happens and the time in which an infected individual infects a susceptible one via an active edge follow an exponential distribution of parameters $\delta = 1$, γ and λ , respectively. Therefore, the epidemic will be a composition of independent Poisson stochastic process [5]. In this type of processes, the probability of having n events in time t follows a Poisson distribution of parameter Λt . Equivalently, the time elapsed between two consecutive events

follows an exponential distribution of probability:

$$P(T_n - T_{n-1} > t) = e^{-\Lambda t}$$

where T_n is the time at which the n th event occurs and is also a random variable. In this case, $\Lambda = \rho(t) + E(t)\lambda + \rho(t)\gamma$, where ρ is the number of infected individuals, E is the number of edges connecting infected and susceptible individuals and λ and γ are the rates of infection and mutation, respectively. Once a random event occurs, this will be a recovery with probability $\frac{\rho}{\Lambda}$, an infection with probability $\frac{E\lambda}{\Lambda}$ or a mutation with probability $\frac{\rho\gamma}{\Lambda}$.

So, at each step of the simulation we generate a random number following an exponential distribution of parameter Λ . This will be the time gap between two consecutive events. Then, we generate another random number x between 0 and 1 following a uniform distribution. If $x < \frac{E\lambda}{\Lambda}$, we will choose a random edge from E and consider that the infection propagates from the infected individual to the susceptible one that form this edge. If $\frac{E\lambda}{\Lambda} < x < \frac{E\lambda + \rho}{\Lambda}$, we will choose a random infected individual to recover. Finally, if $x > \frac{E\lambda + \rho}{\Lambda}$, we will choose a random infected individual to host a virus mutation, creating a new strain to which every individual is now susceptible. In all cases, we need to consider the probability of active edges becoming inactive and vice versa, and do the pertinent changes.

We have used a real-world complex network, consisting of the relationships between users of the Slovak social network Pokec [6]. We have represented each user as a node and considered that an edge connects two nodes if the two users follow each other in Pokec. So after cleaning the data we obtain a complex network consisting of 1,326,208 nodes and 8,585,217 edges, which we believe reflects with great fidelity the contacts between a macroscopic fraction of a population.

III. PHASE DIAGRAM OF A MUTANT EPIDEMIC

Since in this section we need to run thousands of simulations, we will work with random subgraphs of the original network that we have used in the previous section in order to reduce computing time. We will compute the phase diagram for two subgraphs of different size and compare them to see if our suppositions were right. This subgraphs will still be complex networks. The first one, which we will call G_1 , will have 82,363 nodes and 355,875 edges and the second, G_2 , 43,305 nodes and 129,283 edges.

We have seen that, depending on the mutation rate of the virus, an epidemic outbreak can be viewed as an SIR model with an eventual extinguishment of the virus, or as an SIS model, with an endemic phase. It would be useful to know the value of γ for which this transition from one model to another happens, as it would help us predict the evolution of an epidemic.

It is reasonable to suppose that this value, which we will call γ_c , depends on the value of λ , since this affects the total number of infected individuals during the epidemic (see equation (6)), and the number of mutations is proportional to this quantity. We will also assume that $\lim_{\lambda \rightarrow \lambda_c^+} \gamma_c = \infty$, because if the virus does not infect a macroscopic fraction of the population before dying out, it cannot arrive to an endemic phase. So γ_c will be a function that decreases with $\frac{1}{\lambda - \lambda_c}$. Therefore, we will work in a two-dimensional phase space whose components are of the type $(\lambda, \gamma) \in \mathbb{R}^+ \times \mathbb{R}^+$. We will say that points with $\lambda < \lambda_c$ are in the healthy phase.

In the previous section we have seen that $\lambda_c = \langle k \rangle^{-1}$ for homogeneous uncorrelated networks, but now we are working with a real-world network, which is not homogeneous. There are two ways of finding λ_c : we can plot the average number of recovered individuals in an epidemic outbreak without mutations in function of λ . λ_c will be the value of the parameter for which the virus begins to infect a macroscopic number of the population. It can be identified by a change of the behaviour of the graphic. The other way is by plotting the susceptibility of the total number of recovered individuals, which is defined as

$$\chi = N \frac{\langle R_\infty^2 \rangle - \langle R_\infty \rangle^2}{\langle R_\infty \rangle}$$

In finite systems, a peak in this magnitude indicates the presence of a continuous phase transition, so the position of its peak is a good estimator of λ_c [7]. In figure 2, we plot this two ways of estimating λ_c for both networks. Using this data, we can conclude that $\lambda_{c_1} = 0.0700 \pm 0.0025$ and $\lambda_{c_2} = 0.1200 \pm 0.0025$.

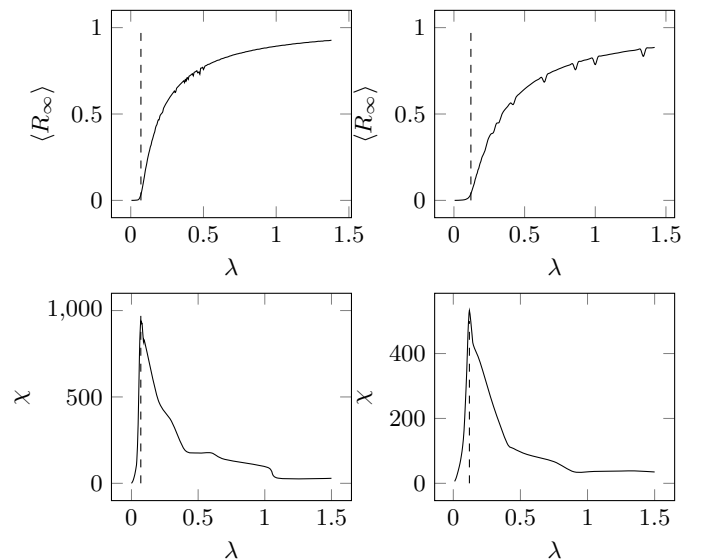


FIG. 2: Graphics showing the average epidemic prevalence (up) and the susceptibility in units of individuals (down) in function of λ , in units of δ . The graphics on the left are for G_1 and the ones in the right for G_2 .

Another assumption we can make is that γ_c is inversely

proportional to the size of the network, because a larger size implies more infected individuals for a same value of λ and, therefore, a bigger probability for the virus to mutate. So in this section we will work with $N\gamma$ instead of γ , and we will try to find a relation $N\gamma_c(\lambda)$ valid for any network size.

We are working with stochastic processes, so when simulating an epidemic, the parameters of which correspond to a point (λ, γ) of the phase space, we will not always get the same trajectory. In particular, there will be points for which some simulations give an extinguishing epidemic and others arrive to an endemic phase. For this points, if we run a large number of simulations, we can compute the probability of arriving to an endemic phase, which from now on we will refer as P , as the fraction of attempts that arrive to a certain time without extinguishing. So we have to consider a time large enough to ensure that an epidemic that arrives alive at this point never extinguishes. In our case, to ensure this we have calculated P for $t = 100$, which is much bigger than the time in which a non-mutant epidemic extinguishes in this networks, and also P for $t = 200$, and we have checked that $P(t = 100) \simeq P(t = 200)$. This means that the epidemics that arrive to $t = 100$, are already on an endemic phase and will not extinguish. So $t = 100$ is a good limit to calculate P .

For a given λ , if we plot P in function of γ , we can distinguish three regions: for γ small enough, $P = 0$ because there are not enough mutations to keep the virus alive forever. On the other hand, for big values of γ , there will be so many mutations that the epidemic arrives to an endemic phase with probability 1. Between these two regions, there is another where P grows linearly with γ . We define γ_c as the smallest value for which $P \neq 0$. If we run enough simulations using different values of γ in this region, we will have a set of points to which we can apply a linear regression. Then, γ_c will be the value for which this straight intersects with the axis $P = 0$ (see Appendix A).

Repeating these process with diverse values of λ will give us a set of points (λ, γ_c) , which we can adjust to obtain the relation $N\gamma_c(\lambda)$ that we are searching. We will do this in the two subgraphs that we have generated to check if this relation is satisfied for any population size.

In addition, we can expect an asymptotic behaviour of γ_c when $\lambda \rightarrow \infty$. That is because the probability of having a mutation is proportional to $N\gamma\rho$, and ρ has an upper bound given by 1, that will be reached for λ large enough. Therefore, for big values of λ the number of mutations will stop growing with λ and γ_c will acquire a constant value. To approximate this value, we have considered that the probability of having a mutation in an infected individual before its recovery is given by $\frac{\gamma}{\gamma+\delta}$, since these two possibilities can be considered as two independent Poisson processes [5]. Then, the estimated number of mutations before the extinction of the virus will be $N\frac{\gamma}{\gamma+\delta}$. In order for the epidemic to survive, this

number must be at least 1, so that this new strain infects all the population almost instantaneously, since $\lambda \sim \infty$, and the whole process repeats forever. This gives us an approximation for $N\gamma_c(\infty)$:

$$N \frac{\gamma_c(\infty)}{\delta + \gamma_c(\infty)} \gtrsim 1 \implies (N-1)\gamma_c(\infty) \simeq N\gamma_c(\infty) \gtrsim \delta = 1$$

We have run simulations with $\lambda = 1,000,000$ δ , an incredibly high value for which the whole population rapidly becomes infected, and have taken $\gamma_c(\infty) \simeq \gamma_c(1,000,000)$. In all the cases, we have obtained a value of $N\gamma(1,000,000)$ very close to $\sqrt{2}$, so we can consider that this hypothesis is correct.

Putting together all the suppositions we have made, we get that the phase diagram of an epidemic will be defined by a curve that will have the following form:

$$\gamma_c = \frac{A}{N(\lambda - \lambda_c)^B} + \gamma_c(\infty) \quad (9)$$

So, if our assumptions and the simulations are correct, we will obtain a set of points $(N\gamma_c, \lambda)$ of the phase space, which we can transform into a set of points of the form $(\log(N(\gamma - \gamma_c)), \log(\lambda - \lambda_c))$, that will define a straight line with slope B and y -intercept $\log(A)$. Then, by making a linear regression of this set of points we can obtain the parameters that are missing in equation 9. In the four cases that we are studying, we obtain equations for the straights that are very close to $y = -x$ (see figure 3).

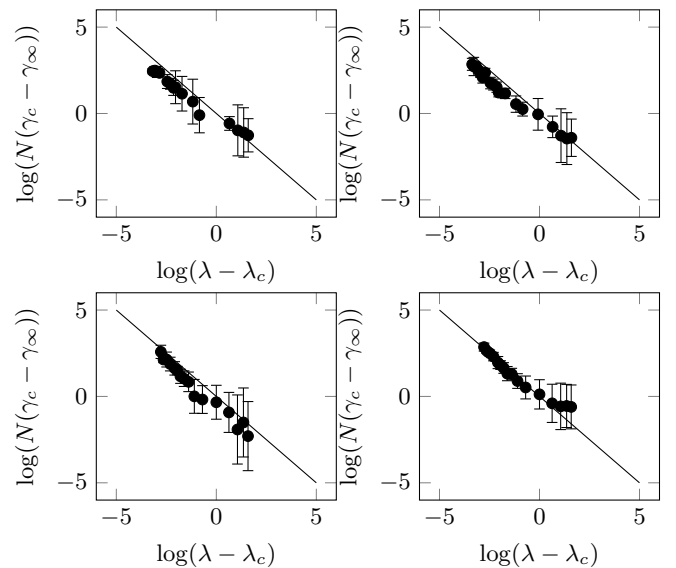


FIG. 3: Plot of the points obtained in the simulations together with the straight $y = -x$. On top, we have the results for G_1 , taking $t = 100$ as the limit on the left and $t = 200$ on the right, both in units of δ^{-1} . On the bottom we have the same for G_2 .

This means that the equation for the critic values of the parameter of mutation is:

$$\gamma_c = \frac{1}{N(\lambda - \lambda_c)} + \gamma_c(\infty) \quad (10)$$

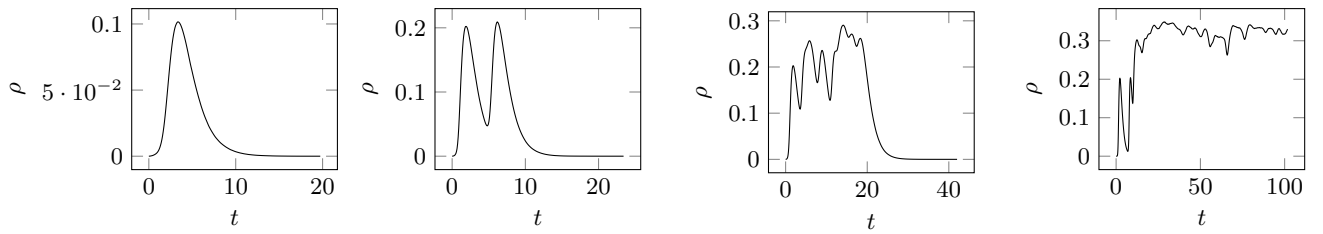


FIG. 4: Time evolution of the fraction of infected individuals by an epidemic in the Pokec network with $\lambda = 0.2 \delta$ for different values of the mutation rate γ . From left to right we have $\gamma = 0$, $\gamma = \frac{\gamma_c}{2}$, $\gamma = \gamma_c$ and $\gamma = 2\gamma_c$, in units of δ . Time is in units of δ^{-1}

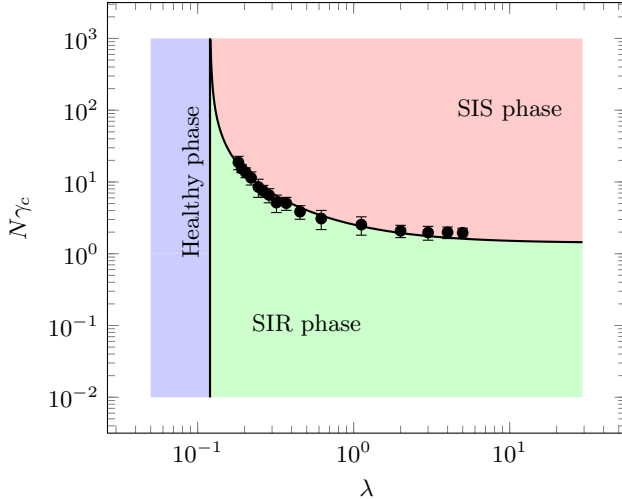


FIG. 5: Phase diagram of an epidemic outbreak in G_2 , given by the equations $N\gamma_c = \frac{1}{\lambda - \lambda_c} + \gamma_c(\infty)$ and $\lambda = \lambda_c$, together with the values obtained in the simulations, in units of δ . In different colours, we can see the three possible behaviours of an epidemic. Both axes are in logarithmic scale.

We can see the form that the phase diagram of an epidemic will have in figure 5. We will check the correctness of equation (10) in the original Pokec network by running four simulations for the same value of λ and different values of γ to see if the behaviour adjusts to the one we have

predicted. In figure 4, we have plotted the evolution of an epidemic outbreak caused by a virus with mutation rate $\gamma = 0$, $\gamma = \frac{\gamma_c}{2}$, $\gamma = \gamma_c$ and $\gamma = 2\gamma_c$, respectively. It can be seen that the evolution of the epidemic goes from a pure SIR model to an SIS model, as we have predicted.

IV. CONCLUSIONS

The time evolution of an epidemic depends strongly on the virus' mutation rate. For any infection rate $\lambda > \lambda_c$, there exists a threshold value γ_c such that for $\gamma \leq \gamma_c$ the epidemic will surely extinguish, and for $\gamma > \gamma_c$, the epidemic can arrive to an endemic phase, and this probability grows with γ . Using computer simulations, we can find that the relation of this critic value γ_c with λ is given by equation (10), with $\gamma_c(\infty) \sim \frac{\sqrt{2}}{N}$. Therefore, for any population, we can plot a phase diagram that will help us predict the evolution of an epidemic in terms of its rates of infection and mutation.

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Appendix A: HOW TO CALCULATE γ_c FOR A GIVEN λ

As we have explained in section III, the graphic of $P(N\gamma)$ has three regions with different behaviour. We can appreciate it on figure 6. Now, we can fit a linear regression to the points in the linear region using the method of least squares [8]. So we will obtain straights with equation $P = AN\gamma + B$. Since we are searching for the minimum value of γ for which $P \neq 0$, we have that

$$N\gamma_c = -\frac{B}{A}$$

In the particular case of figure 9, we get that $A = 0.158 \pm 0.014$ and $B = -0.76 \pm 0.10$, so the value of the critical mutation rate for $\lambda = 0.2$ is

$$N\gamma_c(0.2) = 4.8 \pm 0.8$$

Repeating these steps for several values of λ we get enough points to obtain an equation for the phase diagram of the epidemic.

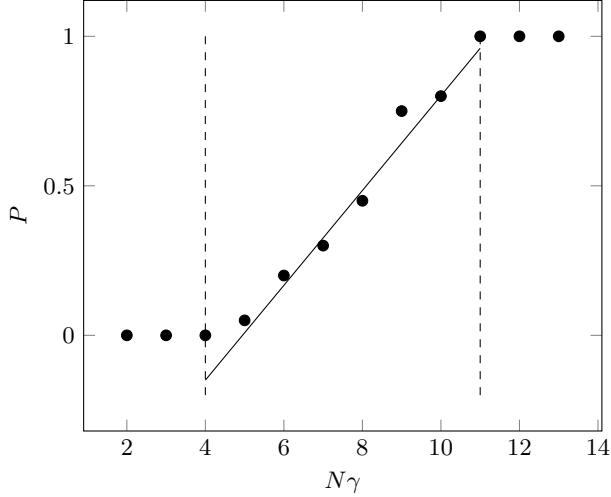


FIG. 6: Phase diagram of G_1 for an infection rate $\lambda = 0.2\delta$, together with the straight $P = 0.16N\gamma - 0.78$, given by the linear regression of the points between the dotted lines. The x axis represents the mutation rate multiplied by the size of the network in units of δ . At the y axis we have the fraction of simulations that stay alive until $t=200$.